

1. Administrative & Study Identification

Study Title: Study of Lenvatinib (MK-7902/E7080) in Combination With Pembrolizumab (MK-3475) Versus Standard of Care in Participants With Metastatic Colorectal Cancer (MK-7902-017/E7080-G000-325/LEAP-017) **Official Title:** A Phase 3 Randomized Study of Lenvatinib in Combination With Pembrolizumab Versus Standard of Care in Participants With Metastatic Colorectal Cancer Who Have Received and Progressed On or After or Became Intolerant to Prior Treatment **Short Title/Acronym:** LEAP-017 **Protocol Number:** MK-7902-017 / E7080-G000-325 **NCT Number:** NCT04776148 **Posting ID:** 1032571238407904511 **Trial Status:** COMPLETED **Sponsor/Company:** Merck (Merck Sharp & Dohme LLC, in collaboration with Eisai Inc.) **Investigational Product:** Lenvatinib (LENVIMA®) and Pembrolizumab (KEYTRUDA®) **Phase of Development:** Phase III **Medical Monitor:** Sponsor Clinical Director / Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess whether lenvatinib plus pembrolizumab is superior to standard of care (regorafenib or TAS-102) with respect to overall survival (OS) in participants with metastatic colorectal cancer. **Secondary Objectives:** To evaluate Progression-Free Survival (PFS), Objective Response Rate (ORR), and Duration of Response (DOR) per RECIST v1.1.

2.2 Background & Rationale Treatment options are highly limited for patients with previously treated metastatic colorectal cancer (mCRC) that is mismatch repair proficient (pMMR) or not microsatellite instability-high (MSI-H). Standard of care therapies such as regorafenib or trifluridine/tipiracil (TAS-102) have limited clinical activity, generally resulting in a median overall survival of approximately 7.0 months. The combination of the multikinase inhibitor lenvatinib and the anti-PD-1 therapy pembrolizumab aims to modulate the tumor microenvironment to increase the immune attack on cancers, hypothesizing a superior overall survival benefit compared to standard of care.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Standard of Care: Regorafenib or TAS-102) **Blinding:** Open-label **Randomization:** 1:1, Parallel Assignment, stratified by the presence or absence of liver metastases.

3.2 Planned Enrollment & Duration Target **\$N\$:** 563 patients **Number of Sites:** 91 medical centers globally **Estimated Study Start:** 08-Apr-2021 **Data Cut-offs for Final Analysis:** - **Global Cohort:** 20-Feb-2023 - **China Cohort:** 27-Sep-2024

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Histologically or cytologically confirmed diagnosis of unresectable and metastatic colorectal adenocarcinoma (stage IV A-C) that is pMMR or not MSI-H by local testing. 3. Measurable disease per RECIST 1.1; must have received and progressed on or after, or became intolerant to prior standard treatment.

4.2 Exclusion Criteria 1. Has a tumor that is microsatellite instability-high (MSI-H) / mismatch repair deficient (dMMR) per local testing. 2. Has a gastrointestinal condition (e.g., malabsorption) that might affect the absorption of the oral study drug. 3. Present or progressive accumulation of pleural, ascitic, or pericardial fluid requiring drainage or diuretic drugs within 2 weeks prior to enrollment. 4. Radiographic evidence of encasement or invasion of a major blood vessel or intratumoral cavitation.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Lenvatinib 20 mg orally once daily + Pembrolizumab 400 mg intravenously once every 6 weeks. (Standard of Care arm receives Regorafenib 160 mg orally once daily Days 1-21 of a 4-week cycle OR TAS-102 [trifluridine and tipiracil hydrochloride; trifluridine trade names: Viroptic, Lonsurf] 35 mg/m² orally twice daily Days 1-5 and 8-12 of a 4-week cycle). **Route of Administration:** Oral (Lenvatinib, Regorafenib, TAS-102/trifluridine-tipiracil) and Intravenous (Pembrolizumab). **Duration of Treatment:** Pembrolizumab for up to 35 cycles (approximately 2 years). Lenvatinib may be administered as a single agent continuously after the 2 years until protocol-specified discontinuation criteria (e.g., disease progression or unacceptable toxicity) are met.

5.2 Concomitant & Prohibited Therapy Permitted: Standard concomitant supportive care medications. **Prohibited:** Concurrent investigational agents, live/live-attenuated vaccines, and specified systemic immunosuppressive therapies during the study period.

6. Study Timeline & Visit Schedule

Visit ID	Window	Key Activities/Procedures	:---	:---
:---	Screening	Days -28 to 0	Informed Consent,	Medical

History, Baseline Tumor Imaging, Eligibility Labs | | **Baseline** | Day 0 | Randomization, First Dose of Study Drug | | **Interim** | Every 4 to 6 Weeks | Safety Labs, Efficacy Assessment, Tumor Imaging per RECIST 1.1 (Q6W for Investigational Arm, Q4W for SOC Arm) | | **End of Study** | Follow-up | Treatment Discontinuation Visit, Safety Follow-Up, Survival Follow-Up |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Overall Survival (OS), defined as the time from randomization to death due to any cause.

Measurement Method: Clinical tracking and survival follow-up evaluated independently for the following cohorts: - **Global**

Cohort: Final Analysis performed with a data cut-off date of 20-Feb-2023. - **China Cohort:** Final Analysis performed with a data cut-off date of 27-Sep-2024.

7.2 Secondary & Exploratory Endpoints - Progression-Free Survival (PFS) assessed by Blinded Independent Central Review (BICR) per modified RECIST 1.1 (Analyzed for both Global and China Cohorts). - Objective Response Rate (ORR) assessed by BICR per modified RECIST 1.1 (Global Cohort). - Duration of Response (DOR).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Collection of treatment-related adverse events and all-cause AEs monitored continuously throughout the treatment and follow-up periods. **Serious Adverse Events (SAEs):** Evaluated strictly and typically reported within 24 hours to the sponsor. **Data Monitoring Committee (DMC):** An external data safety monitoring committee reviewed interim study results to ensure patient safety and to assess efficacy at prespecified interim boundaries.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intention-to-Treat (ITT) population for primary efficacy outcomes. The Global Cohort was evaluated for efficacy separately from the China Cohort. Per the supplemental Statistical Analysis Plan (sSAP), China participants randomized after the enrollment of the global portion closed were not included in the global analysis populations. **Sample Size**

Justification: The trial was powered to test the superiority of the lenvatinib + pembrolizumab combination over the SOC arm utilizing a prespecified statistical significance threshold ($P = 0.0214$). **Handling of Missing Data:** The Cox proportional hazard model with Efron's method of tie handling was used to assess the magnitude of the treatment difference between arms.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study was conducted in accordance with Good Clinical Practice (GCP) guidelines and the Declaration of Helsinki. **Monitoring Plan:** Routine scheduled onsite and remote monitoring visits by the sponsor. **Audit Trail:** Blinded Independent Central Review (BICR) to handle all tumor imaging assessments to prevent bias, along with standard eCRF tracking and data clarification processes.

11. Study Identifiers & Governance

Primary ID: MK-7902-017 / E7080-G000-325 **Registry Number:** NCT04776148 **Posting ID:** 1032571238407904511 **Trial Status:** COMPLETED **Sponsor/Lead Organization:** Merck **Official Regulatory Title:** A Phase 3 Randomized Study of Lenvatinib in Combination With Pembrolizumab Versus Standard of Care in Participants With Metastatic Colorectal Cancer Who Have Received and Progressed On or After or Became Intolerant to Prior Treatment

12. Clinical Framework (mCRC Specific)

Disease Area: Colorectal Neoplasms **Preferred UMLS Name:** Metastatic Colorectal Cancer **Diagnosis Threshold:** Histologically/cytologically confirmed unresectable/metastatic pMMR or not MSI-H colorectal adenocarcinoma **Medical Methodology:** Lenvatinib plus Pembrolizumab combination therapy vs. Standard of Care (Regorafenib or TAS-102) **Optimization Target:** Overall Survival (OS) and objective tumor response

13. Study Procedures & Methodology

Management Pathway: 1:1 Randomization to Investigational Arm or Standard of Care **Education Protocol (HCP):** Investigator site training on specific dosing, modification regimens, and adverse event management for multikinase inhibitors and immune checkpoint inhibitors **Education Protocol (Patient):** Informed consent processing, AE reporting education, and clinic visit scheduling **Quality Audit Mechanism:** Blinded Independent Central Review (BICR) for objective assessment of tumor progression and overall response

14. Observation & Follow-Up Schedule

Total Duration: Up to approximately 2 years for Pembrolizumab; Lenvatinib continuous until progression or toxicity **Onsite Visit Cadence:** Every 4 to 6 weeks aligning with treatment cycles **Remote Monitoring Frequency:** Telephone or virtual visits incorporated

continuously between cycles **Checklist Review Interval:** Regular tumor imaging intervals as prespecified in the trial protocol to assess PFS and ORR

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				